

AI Agents :

From Language Generation to Task Automation

an essay in AI engineering history and epistemology

Louis Jourdain

LibreOffice



Who am I ?

Louis Jourdain

AI engineer (NLP)

Paris, Île-de-France, France · **Coordonnées**

Plus de 500 relations

Articles et actes de colloques

- **TALN 2025, Marseille**, Atelier EvalLLM (DGA)
  Générer pour mieux tester : vers des datasets diversifiés pour une évaluation fiable des systèmes de Question Answering
  Louis Jourdain, Skander Hellal
- **CAID Rennes 2024**
  Pourquoi se limiter à une recherche quand on peut l'étendre ? Amélioration de l'architecture RAG par des stratégies d'expansion de requêtes et d'agrégation de documents
  Louis Jourdain, Skander Hellal, Tony Marini
  hal.science/hal-05046658v1
- **Journée d'étude sur le TAL frugal et la RI frugale** (Sorbonne, organisée par Gaël Lejeune et Caio Corro)
  Essai d'étude quantitative des schémas de composition nominale en grec ancien : la quête de données de qualité
  PDF
- **CAID 2025, Rennes**
  Hybrid Canonicalization of Intelligence Knowledge Graphs: A Frugal and Explainable Approach
  Louis Jourdain, Pauline Degez

Autres communications

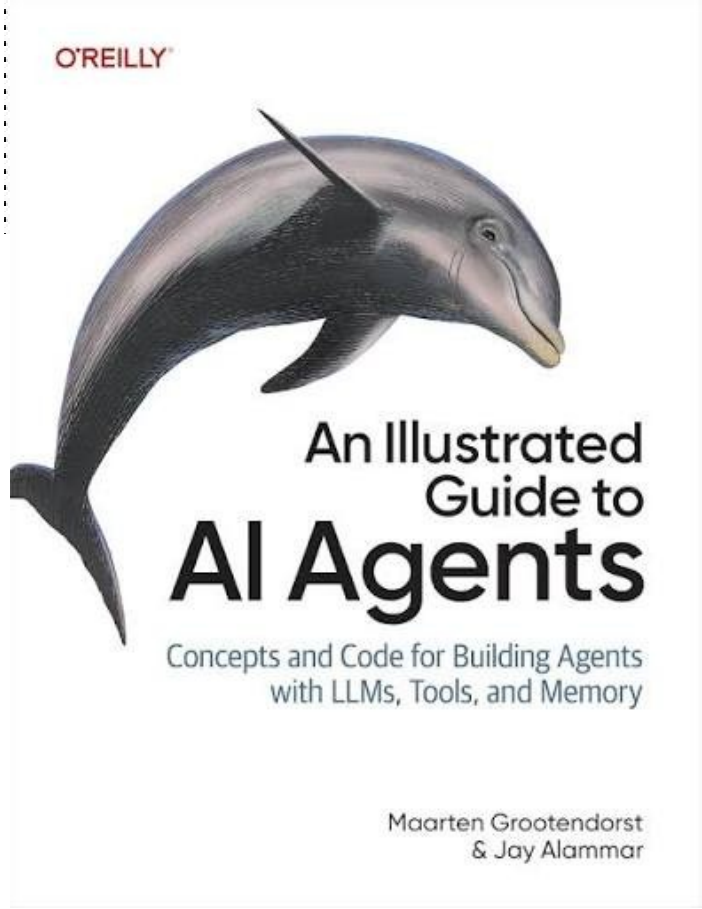
- **Meetup sur les Graphes de connaissance**
  Expansion de requête et Graphes de Connaissance comme support du RAG
  YouTube
- **Meetup Elastic #93**
  Traversée d'un graphe, avec des millions de relations, pour améliorer un RAG
  YouTube
- **Livre Blanc du Pôle d'Excellence Cyber – "IA de confiance dans la Défense"**
  Les principales vulnérabilités des LLMs face aux attaques malveillantes
  Contribution avec *Christian Langevin*

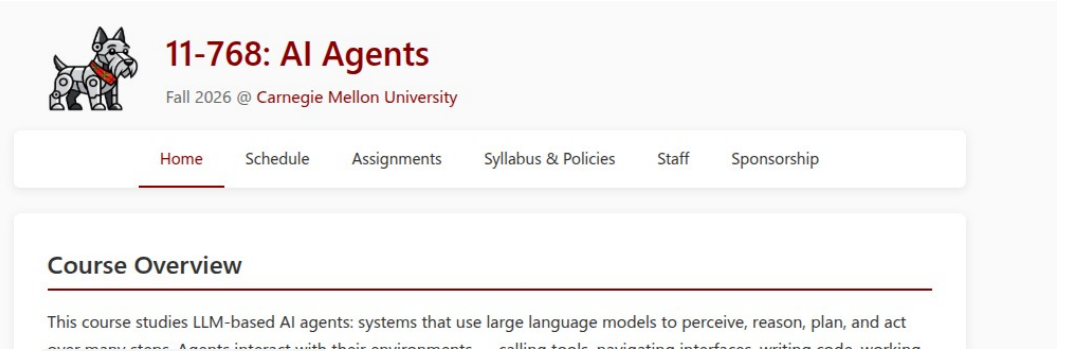
  Livre Blanc

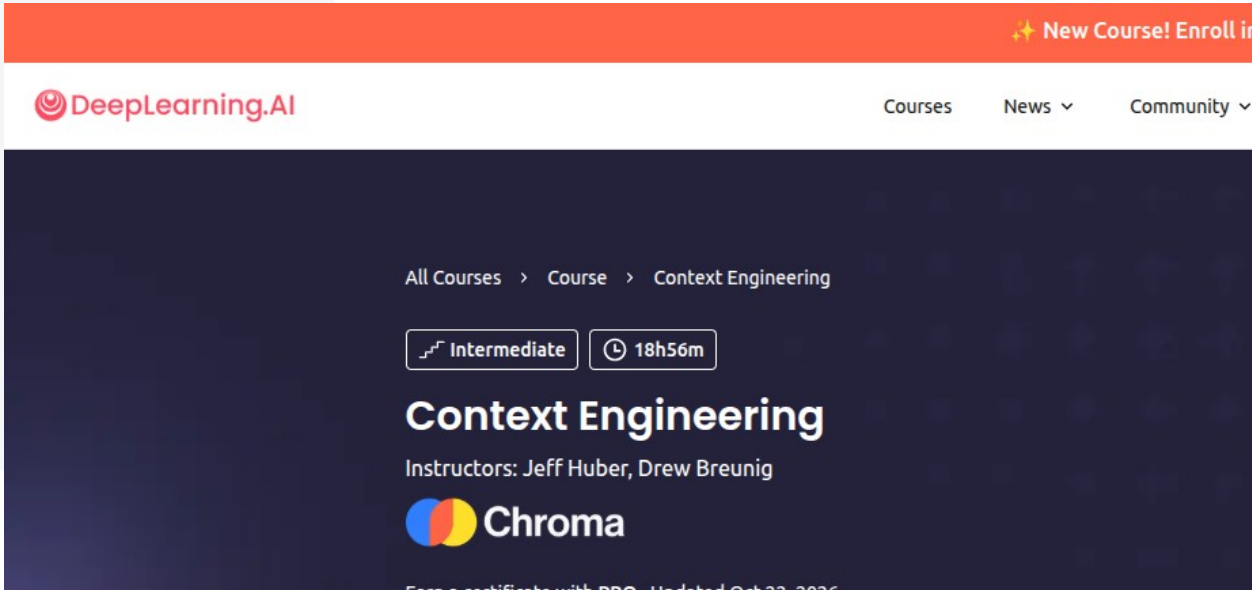
For this course

- 3b13jourdainl@gmail.com
- https://github.com/Dinarque/AI_Agents_Paris_Cite/tree/main

Recommended Extra material








WE ARE AN "AGENTIC AI" COMPANY
BE HONEST
I AM BEING HONEST
HOW DO YOU DEFINE "AI AGENTS"?
FANCY CHATBOTS AND COPILOTS
THANK YOU


UMBRELLA
$9,95
AGENTIC AI UMBRELLA
$999,95
@ordax
eordax

What do you expect from the class ?

6 Weeks of intesive Agentic Activity

Chapter 1
What are AI Agents

Chapter 2
Expanding LLM capacities with Tools

Chapter 3 :
Conditionning behaviour Memory and Context

Chapter 4
Architectures and Engineering

Chapter 5
Test yourself
Real use cases + Demo

Chapter 6
Demo

What you expect : Evaluation

Maybe
Written exam on week 5 :

Content of the first 4 weeks
Question on course,
Code,
Practical use case
On Paper, no notes

Group project,
2 or 3 students

Implement an Agent
or Agentic system to solve a
Domain specific issue.
Deadline 30/05
Several topics available

What are all of the LinkedIn AI buzzwords of 2026 ?

https://neurips.cc/Downloads/2026

The 2025 AI Agent Index Documenting Technical and Safety Features of Deployed Agentic AI Systems

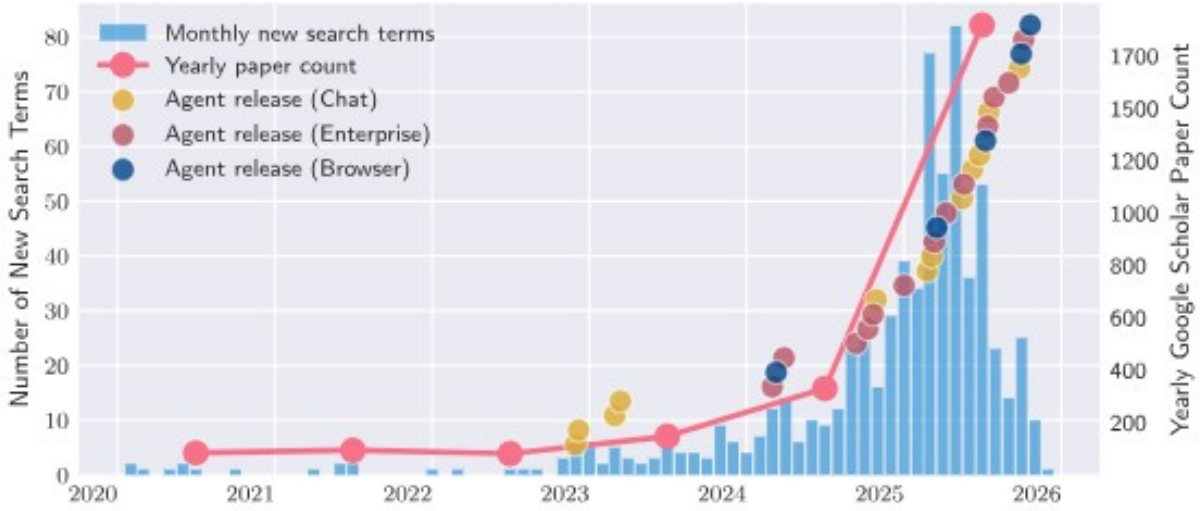
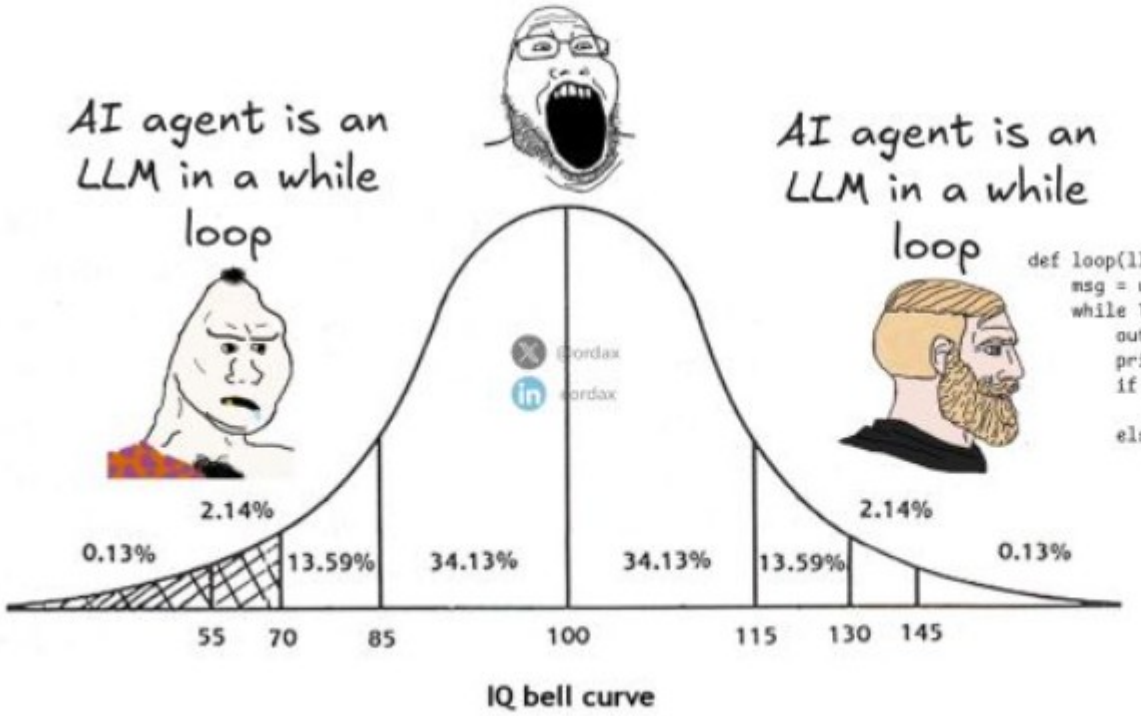


Figure 1. **Interest in AI agents is growing.** 2025 has seen a sharp increase in interest in AI agents. This is reflected in an

What do they mean by AI Agent?

An AI agent is a software system that autonomously perceives its environment, makes decisions, and takes actions to achieve specific goals, often learning and adapting over time




```
def loop(llm):
    msg = user_input()
    while True:
        output, tool_calls = llm(msg)
        print("Agent: ", output)
        if tool_calls:
            msg = [ handle_tool_call(tc) for tc in tool_calls ]
        else:
            msg = user_input()
```



Data + Machine Learning =





Data + Artificial Intellifence =





Data + Generative AI =

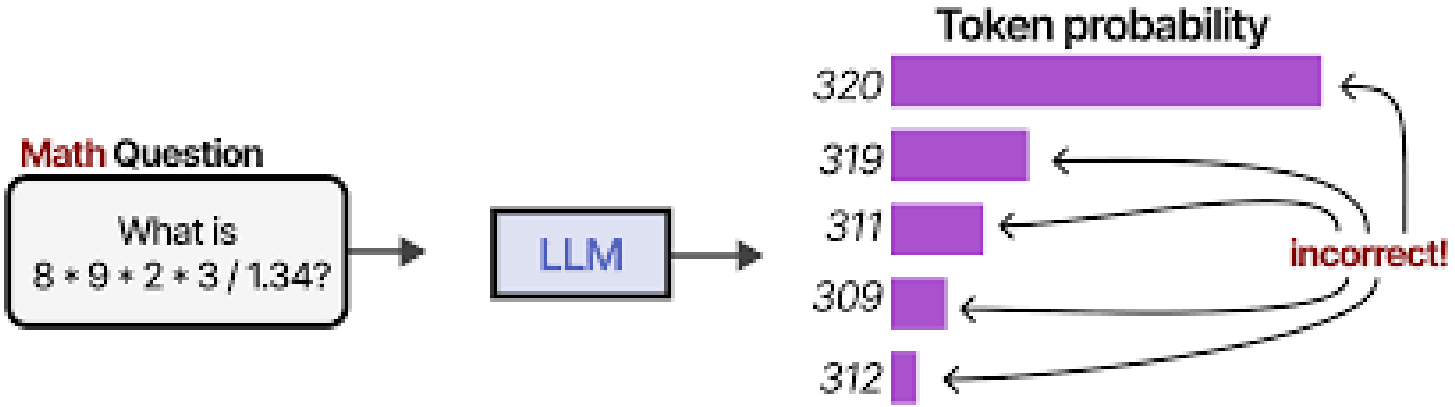




Data + Agentic AI =



At the beginning, there was the LLM



This illustration and the ones in the same style all come from the EXCELLENT Blog of https://newsletter.maartengrootendorst.com/

For the memes, see Eduardo Ordax on LinkedIn

From LLMs to Agents

LLMs predict the next word → Agents automate complex tasks



2023: Research Wave
Heterogeneous explorations



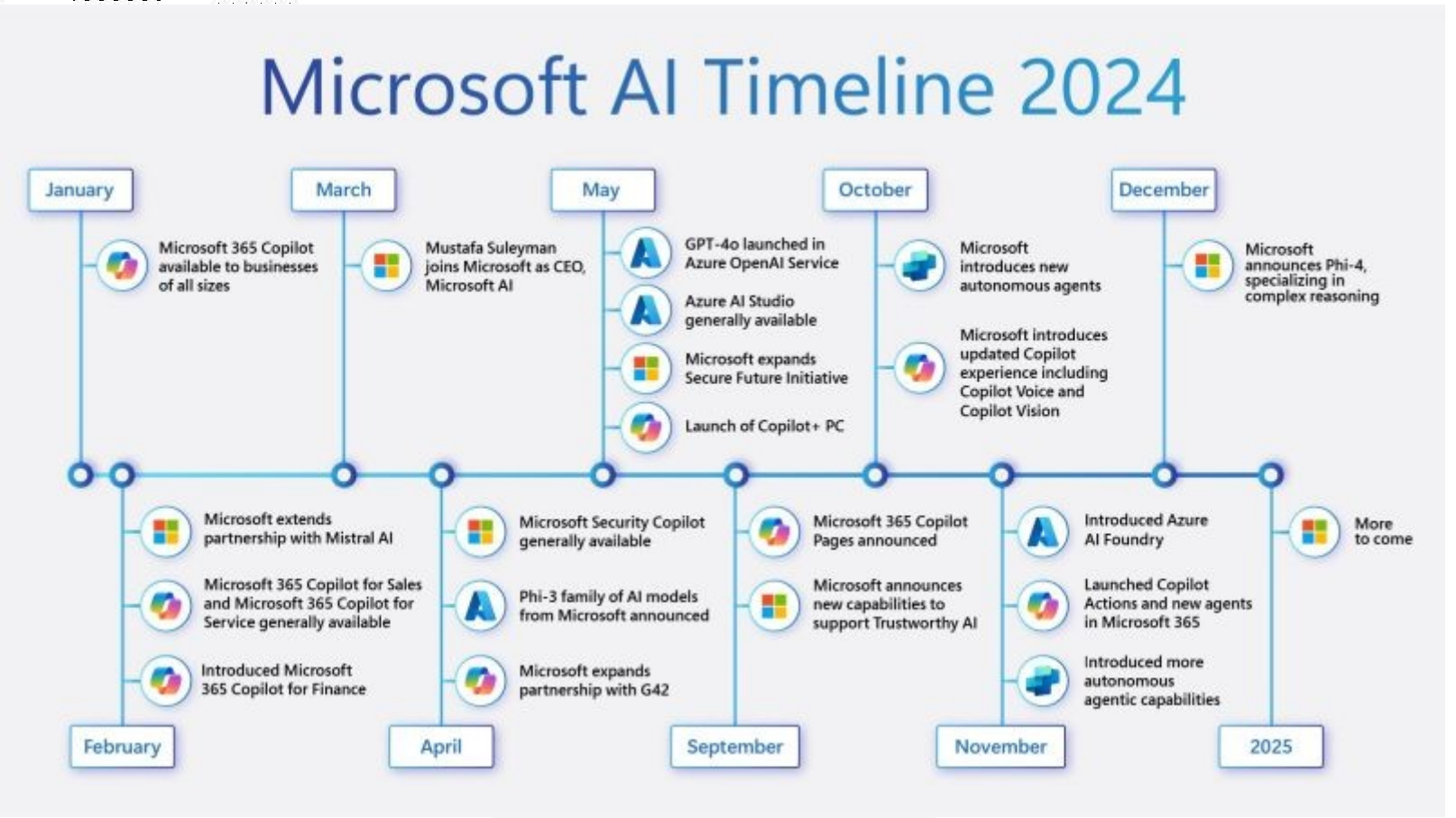
2024: Stabilization
Clear definitions emerge



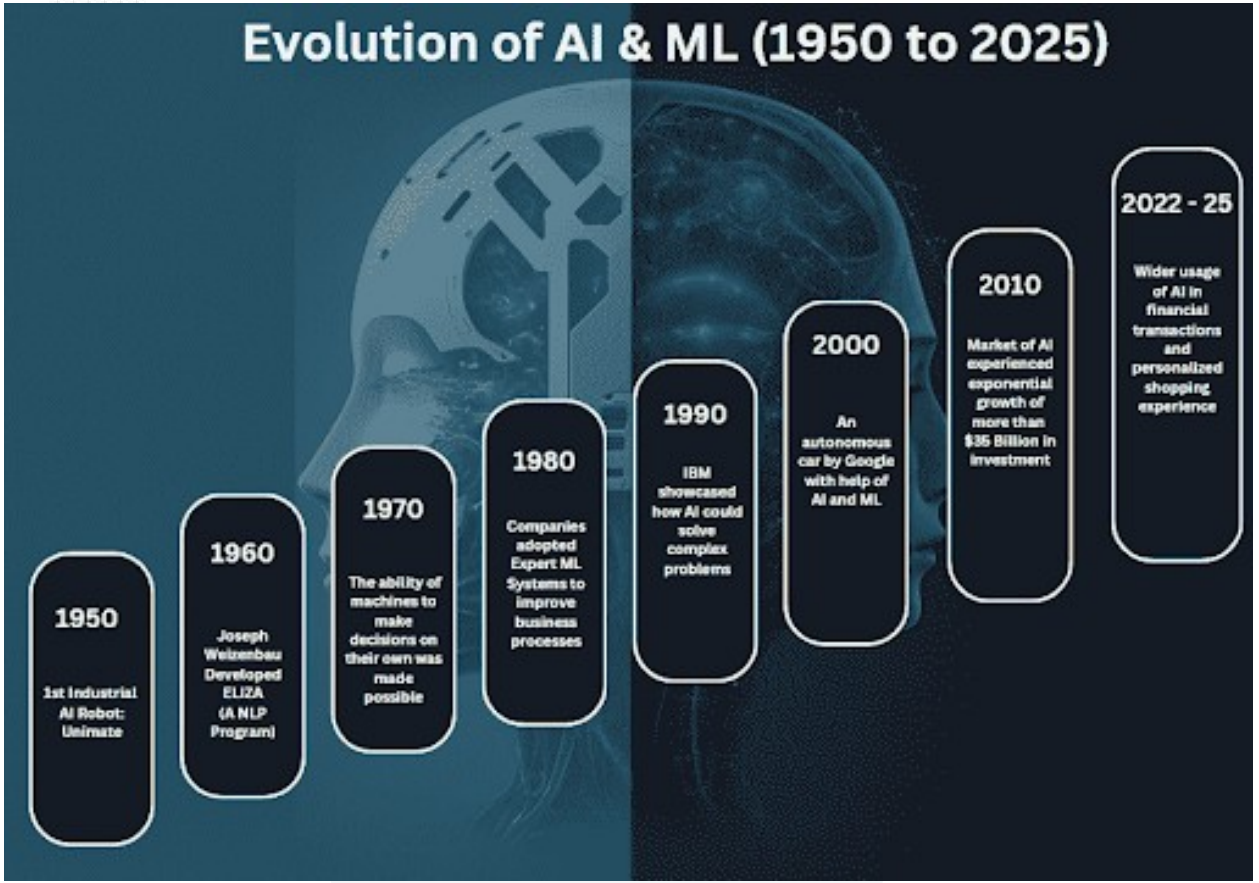
2025: Production
Industrialization begins



Challenge
Building on unfinished GenAI adoption



https://www.linkedin.com/posts/dipendra-shekhawat_2025-would-be-more-on-ai-agents-agentic-activity-7275708243501297664-5ogI



https://internetsearchinc.com/ai-agents-the-next-frontier-in-technology-for-2025/

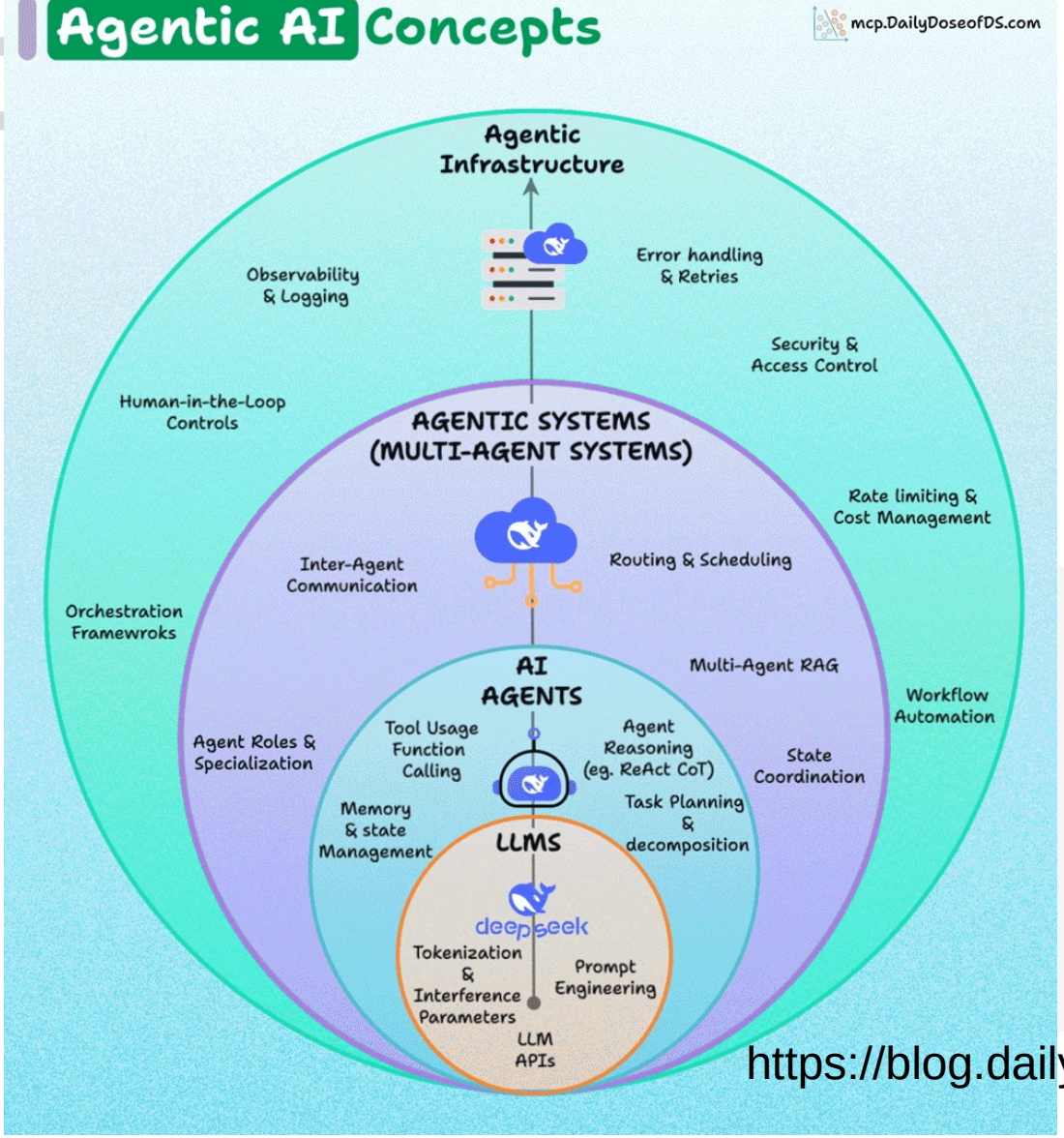
What characterizes a technological revolution ?

> *"Without the hammer, the carpenter cannot yet build the house"*

✅ **Entry Level**
Any Python developer can build a functional agent

⚙️ **Production Scale**
Requires: robust infrastructure, dedicated data teams, governance systems

🏢 **Major Players**
Microsoft, Google, IBM, Amazon, Salesforce building agentic ecosystems



https://blog.dailydoseofds.com/p/4-layers-of-agentic-ai-syste

In Academia : the Paper Frenzy

- *Building crypto portfolios with agentic AI*
- *AI Agents and the Attention Lemons Problem in Two-Sided Ad Markets*
- *AI-Driven Generation of Old English: A Framework for Low-Resource Languages*
- *HAMLET: Hyperadaptive Agent-based Modeling for Live Embodied Theatrics*
- *TacticCraft: Natural Language-Driven Tactical Adaptation for StarCraft II*
- *Synthetic Socratic Debates: Examining Persona Effects on Moral Decision and Persuasion Dynamics*
- *AutoGen Driven Multi-Agent Framework for Iterative Crime Data Analysis and Prediction*

I) AI Agents and Society



$450 Billion

Potential value by 2028

Current Adoption

At scale	**2%**
Partial deployment	**12%**
Pilot stage	**23%**
Exploration	**61%**

By 2027-2028

✅ 50% of business decisions augmented/automated

⚠️ 40% of projects canceled

🤝 40% integrate agents as "team members"

Gartner Predicts Over 40% of Agentic AI Projects

Confidence in Fully Autonomous Systems

43%
Initial confidence

→

27%
One year later

The future is human-AI collaboration, not full replacement

Dario Amodei's Prediction (Anthropic CEO)

AI could eliminate 50% of entry-level white-collar jobs
10-20% unemployment within 1-5 years

Microsoft Study: 40 Most Affected Professions

High Risk

- Interpreters & translators
- Historians
- Sales representatives

Surprisingly Affected

- Data scientists
- Mathematicians

⚠️ Critical Distinction

Current study: GenAI as assistant (human-in-the-loop)
Real disruption: Agentic AI with autonomy & orchestration

Table 3: Top 40 occupations with highest AI applicability score.

Job Title (Abbrv.)	Coverage	Cmpltn.	Scope	Score	Employment
Interpreters and Translators	0.98	0.88	0.57	0.49	51,560
Historians	0.91	0.85	0.56	0.48	3,040
Passenger Attendants	0.80	0.88	0.62	0.47	20,190
Sales Representatives of Services	0.84	0.90	0.57	0.46	1,142,020
Writers and Authors	0.85	0.84	0.60	0.45	49,450
Customer Service Representatives	0.72	0.90	0.59	0.44	2,858,710
CNC Tool Programmers	0.90	0.87	0.53	0.44	28,030
Telephone Operators	0.80	0.86	0.57	0.42	4,600
Ticket Agents and Travel Clerks	0.71	0.90	0.56	0.41	119,270
Broadcast Announcers and Radio DJs	0.74	0.84	0.60	0.41	25,070
Brokerage Clerks	0.74	0.89	0.57	0.41	48,060
Farm and Home Management Educators	0.77	0.91	0.55	0.41	8,110
Telemarketers	0.66	0.89	0.60	0.40	81,580
Concierges	0.70	0.88	0.56	0.40	41,020
Political Scientists	0.77	0.87	0.53	0.39	5,580
News Analysts, Reporters, Journalists	0.81	0.81	0.56	0.39	45,020
Mathematicians	0.91	0.74	0.54	0.39	2,220
Technical Writers	0.83	0.82	0.54	0.38	47,970
Proofreaders and Copy Markers	0.91	0.86	0.49	0.38	5,490
Hosts and Hostesses	0.60	0.90	0.57	0.37	425,020
Editors	0.78	0.82	0.54	0.37	95,700
Business Teachers, Postsecondary	0.70	0.90	0.52	0.37	82,980
Public Relations Specialists	0.63	0.90	0.60	0.36	275,550
Demonstrators and Product Promoters	0.64	0.88	0.53	0.36	50,790
Advertising Sales Agents	0.66	0.90	0.53	0.36	108,100
New Accounts Clerks	0.72	0.87	0.51	0.36	41,180
Statistical Assistants	0.85	0.84	0.49	0.36	7,200
Counter and Rental Clerks	0.62	0.90	0.52	0.36	390,300
Data Scientists	0.77	0.86	0.51	0.36	192,710
Personal Financial Advisors	0.69	0.88	0.52	0.35	272,190
Archivists	0.66	0.88	0.49	0.35	7,150
Economics Teachers, Postsecondary	0.68	0.90	0.51	0.35	12,210
Web Developers	0.73	0.86	0.51	0.35	85,350
Management Analysts	0.68	0.90	0.54	0.35	838,140
Geographers	0.77	0.83	0.48	0.35	1,460
Models	0.64	0.89	0.53	0.35	3,090
Market Research Analysts	0.71	0.90	0.52	0.35	846,370
Public Safety Telecommunicators	0.66	0.88	0.53	0.35	97,820
Switchboard Operators	0.68	0.86	0.52	0.35	43,830
Library Science Teachers, Postsecondary	0.65	0.90	0.51	0.34	4,220

Note: Metrics reported as mean of user goal and AI action score.

🤖 Autonomy

Engineering:
Independent task execution

Moral:
Self-governance & ethical principles

AI agents have engineering autonomy, not moral autonomy

⚖️ Responsibility Gap

Problem:

When AI causes harm, who is accountable?

Proposed Solutions:

- Preventative restrictions
- Shared accountability
- New legal frameworks

💫 The Messianic Stance

Agents framed as revolutionary infrastructure—next after Internet & smartphones

⚠️ Risk: Hype obscures governance, safety, and failure modes

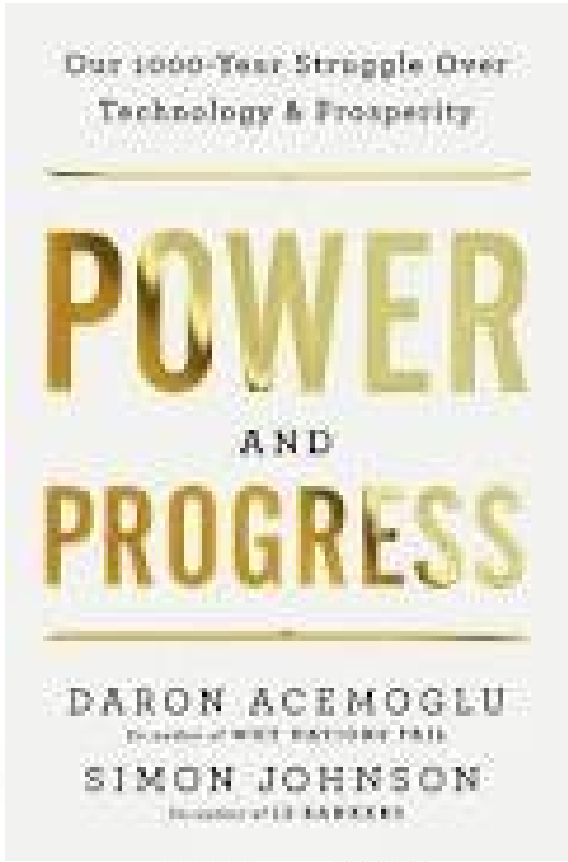
AI, Human Cognition and Knowledge Collapse*

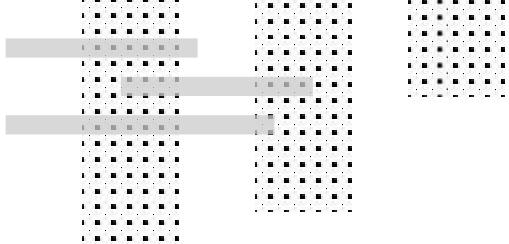
Daron Acemoglu† Dingwen Kong‡ Asuman Ozdaglar§

February 20, 2026

Abstract

We study how generative AI, and in particular agentic AI, shapes human learning incentives and the long-run evolution of society's information ecosystem. We build a dynamic model of





Nicole D'Angelo
B2B content strategist & writer | Content support from ideation to execution
1 mois

Every AI evangelist on here posts like:

👊 The future is here.
🪥 There's no putting the toothpaste back in the tube.
🤖 AI is changing the world. The question is, how will YOU adapt?
🐂 Will you take the bull by the horns?
🐑 Or will you be a sheep?
😟 A lame, Luddite sheep.
💸 But if you master AI, you'll be soaring to new heights. While holding onto the horns of that bull. It's a flying bull now.
🤔 How does this happen?
🔥 By becoming a truly strategic innovator.
🔥 🔥 And by investing in my AI startup that's definitely NOT just something you could do on your own with ChatGPT. Promise.
🔥 🔥 🔥 Those who rise to the new world order will be the masters of it.

[Insert about 50 likes and a dozen AI-generated comments all saying "So true, Chad!"]

The Future of Marketing: From Specialist to Generalist

✦ Ce titre a été résumé par l'IA à partir du post ci-dessous.



Aaron Nava
1 mois

The Specialist is Dead. The "Empty Calories" Generalist Is Next.

AI has democratized skills, which means the traditional marketing specialist is facing a harsh reality. Why hire a copywriter when a skilled prompt engineer can get a dozen outstanding versions (in any style or combination of) in seconds? Why hire an SEO analyst when a single tool can audit a site in minutes? It's clear the value proposition has shifted. The future of marketing belongs to the generalist. That is, the person who can orchestrate these powerful AI tools to build a cohesive, multi-channel strategy.

But I see a problem on the horizon. This is also creating a new kind of marketing professional: the "empty calories" generalist. This is someone who can use the tools but lacks the fundamental understanding of why they are using them. They can write a prompt for a video script, but don't understand marketing strategy, or how to build brands that last. They have a superficial knowledge base, built on AI outputs, not on hands-on experience and deep critical thought.

The winning skills in the new world won't be those who simply know how to use AI. They will be the marketers who have a deep, foundational understanding of marketing principles AND can use AI as a force multiplier. This is the difference between an AI user and an AI commander.

The question is, if AI can generate a thousand ideas, what do you think is the single human skill that proves we found the right one?

What about Muse ?

Le Figaro

Le nouvel agent IA de Meta, Muse, déjà pris en défaut par une faille de sécurité



il y a 16 heures

Siècle Digital

Avec Claude Duet, Anthropic veut concurrencer Meta Muse et OpenAI Dots



il y a 10 heures

VOI.ID

L'application d'IA Meta Muse atteint 10 millions de téléchargements aux États-…



il y a 23 heures

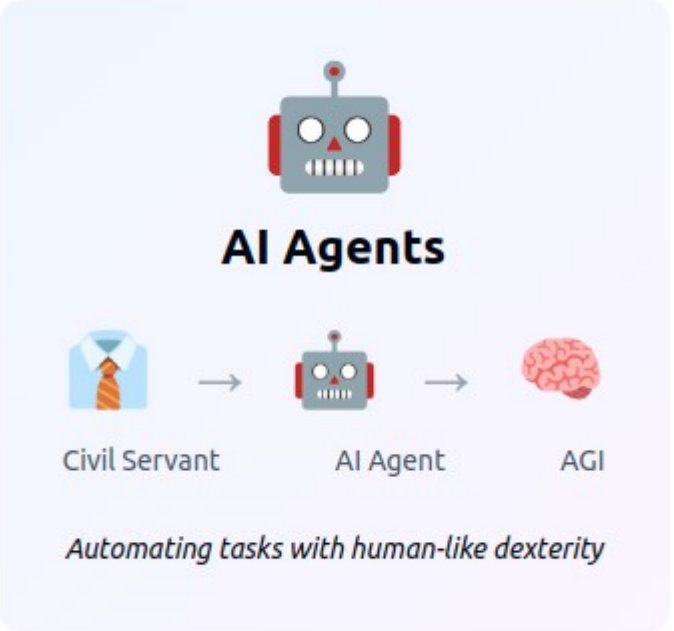
Frandroid

Presque trois fois mieux que ChatGPT à ses débuts : l'agent IA Muse de Meta…



Il y a 1 jour

II) AI Agents, a contemporary definition



Etymology: Latin "agere"

"One who acts, produces, puts into effect"

Medieval philosophy → Administrative → Computer Science (1990s)

🌐 Semantic Web Era

Software dispatched across networks to perform tasks autonomously

Examples: crawlers, bots, mobile agents

🔄 Automata

State machines with predetermined transitions

AI agents: dynamically modify their own structure

🤝 Multi-Agent Systems

Multiple autonomous agents interacting in shared environments

Used in: simulations, economics, gaming (NPCs)

Russell & Norvig:
"Perceiving environment through sensors and acting through actuators"

Hugging Face:
"AI model capable of reasoning, planning, and interacting with environment"

IBM:
"System capable of autonomously performing tasks by designing its workflow and utilizing tools"

Cognizant:
"Goes beyond creating output; acts on information to achieve desired outcomes"

Problem: Definitions are vague and evolving

Solution: Think of agentivity as a spectrum

Agency Level	Description	How that's called	Example Pattern
☆☆☆	LLM output has no impact on program flow	Simple Processor	`process_llm_output(llm_response)`
★☆☆	LLM output determines an if/else switch	Router	`if llm_decision(): path_a() else: path_b()`
★★☆	LLM output determines function execution	Tool Caller	`run_function(llm_chosen_tool, llm_chosen_args)`
★★★	LLM output controls iteration and program continuation	Multi-step Agent	`while llm_should_continue(): execute_next_step()`
★★★	One agentic workflow can start another agentic workflow	Multi-Agent	`if llm_trigger(): execute_agent()`

Fast-Moving Ecosystem

Many libraries, different philosophies

LangChain

Complex tasks, many features, less transparent

LangGraph

Graph-based workflow management

CrewAI

Multi-agent collaboration, hierarchical

smolagents

Simple, interpretable, transparent

AutoGen

Conversational agents, Microsoft

OpenAI Swarm

Lightweight multi-agent

Comparative Table of LLM Agent Frameworks

Framework	Key Characteristics	Advantages	Drawbacks
LangChain / LangGraph	- Python, modular, chain-oriented - LangGraph: graph-based orchestrations - Tools, memory, planning - Multi-LLM compatible	+ Highly flexible + Huge ecosystem (LangSmith, many tools) + Multi-agent possible with LangGraph + Very popular	– Complex to learn – Verbose – Risk of overhead for small projects
Semantic Kernel	- C# and Python - Plugin architecture (skills) - Integrated planner - Strongly oriented toward productivity and enterprise apps	+ Ideal for .NET ecosystem + Microsoft support + Enterprise extensibility	– Less rich than LangChain for RAG or advanced agents – Steeper learning curve for planner
CrewAI	- Python + No-code studio - Role/task orchestration - Sequential orchestration - 700+ API/tool integrations	+ Very productive for MVPs or startups + Integrated UI/monitoring + Good code/no-code balance	– Less flexible for free orchestration – Heavy for simple projects – Telemetry enabled by default
smol-agents (HF)	- Python - Very lightweight (1 script = 1 agent) - Calls Python functions from generated code - HF Transformers integration	+ Extremely simple + Highly customizable + Few dependencies + Great for local or experimental agents	– No native memory – No multi-agent coordination – Less mature – Requires user-provided monitoring and security
OpenAI Swarm	- Python - Agents + handoffs - Stateless execution - Client-side / edge-first design	+ Minimalistic and educational + Highly customizable + Private and secure (no persistence)	– Not suited for production – No long-term memory – Requires supervision and strict error handling
Autogen (Microsoft)	- Python - Asynchronous inter-agent communication - Configurable roles - Native multi-agent support + human-in-the-loop	+ Very powerful for coordination + Open-source with Microsoft support + Scalable async workflows	– High complexity – Limited UI/monitoring unless integrated manually
Google ADK (Agent Development Kit)	- Python/Go - Strong focus on reproducibility and deployment - Artifact system for persistent outputs - Cloud-ready integrations (Vertex AI, GCP services) - Modular plugin system	+ Clean artifact system (manages files, graphs, logs) + Cloud-native orchestration + Good for structured workflows and observability + Designed for production-scale deployment	– Newer ecosystem, less community adoption – Tied to Google Cloud stack – Some features still experimental

⚖️ Trade-off 1: Power vs. Transparency



Simple & Explainable
(smolagents)

↔



Complex & Opaque
(LangChain, CrewAI)

⚖️ Trade-off 2: Ease vs. Control



Easy to Write
Less control

↔



More Control
More code

📝 Return to Simplicity

Anthropic & Hugging Face advocate for readable, understandable code

🎯 Different Objectives

Not all frameworks solve the same problems—choose wisely

🔮 Disruption Ahead

Vision-enabled agents and "computer use" may reshape the landscape

⚠️ No Perfect Solution

Missing: robust memory management, complete observability

🤝 Complementary Tools

Combine frameworks based on project needs

Activity: OpenAI API vs ChatGPT

Key Question:
What's the difference between the raw model and the product?

🔌 OpenAI API

- ✓ Deterministic (temp=0)
- ✓ No memory
- ✓ No default persona
- ✓ Just the model
- ✓ You control everything

💬 ChatGPT

- ✓ Variable responses
- ✓ Automatic memory
- ✓ Built-in personality
- ✓ Code execution
- ✓ Orchestrated workflows

ChatGPT = Raw Model + Agentic Layer

What Users Think vs. Reality

❌ Perception

"I'm talking directly to the AI model"



✅ Reality

"I'm interacting with an agent built around the model"



The scaffolding is invisible, but it's what makes the experience coherent

Production Use: API Calls

1. **Store Prompts**
   Consistent, reproducible requests
2. **Parse Responses**
   Enforce structured output (JSON)
3. **Inject as Variables**
   Feed into next pipeline stage
4. **Manage History**
   Developer's responsibility for conversations

💔 Traditional Programming

Same input → Same output (always)

Predictable, debuggable, trustworthy

↓

🎲 GenAI Programming

Same input → Different outputs (probably)

Probabilistic, harder to debug, needs guardrails

New Developer Skills:

- Prompt engineering to constrain outputs
- Temperature tuning to reduce variability
- Parsing & validation mechanisms
- Working with probabilities, not guarantees

🏢 Traditional Software

Self-contained codebase
Full control over execution

↓

☁️ GenAI-Powered Software

✓ Depends on external API availability

✓ Subject to latency & rate limits

✓ Model versions may change silently

✓ Behavior can shift unpredictably

New Challenge:
Taming instability becomes part of the craft

LLMs as Control Structures

Three Roles of LLMs in Applications



Content Generators

Summaries, drafts, answers



Content Extractors

Parse unstructured text into variables



Decision Makers

Replace if/else and rules engines

LLMs change the texture of programming

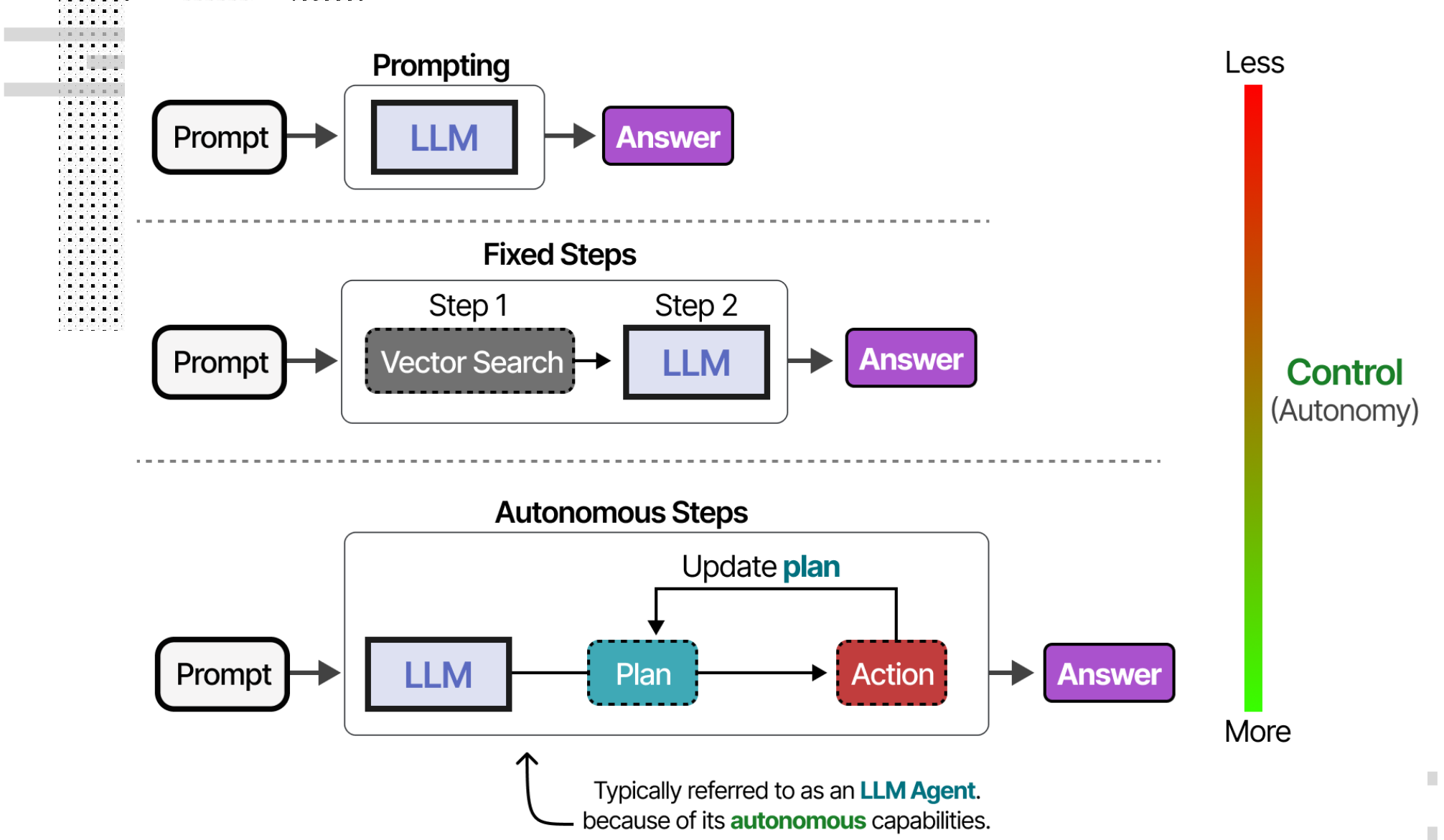
From deterministic functions → Probabilistic orchestration

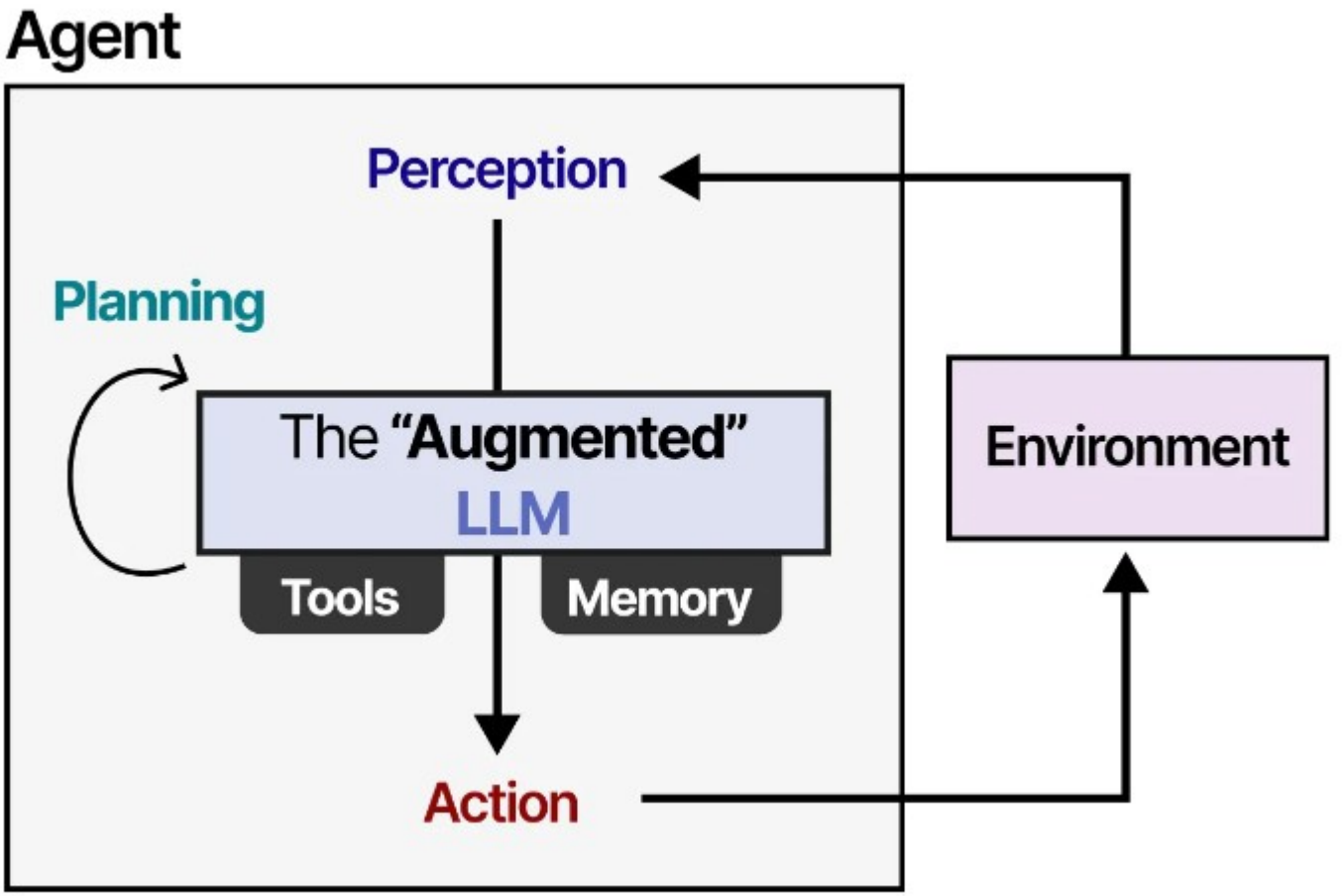


Official Course Definition

"A computer program that has the ability to call a Large Language Model in order to autonomously solve a complex task without external guidance or supervision."

Autonomy | **Complexity** | **No Supervision**


Prompting
Prompt
LLM
Answer
Fixed Steps
Step 1
Step 2
Prompt
Vector Search
LLM
Answer
Autonomous Steps
Update plan
Prompt
LLM
Plan
Action
Answer
Typically referred to as an LLM Agent.
because of its autonomous capabilities.
Less
Control
(Autonomy)
More


Agent
Perception
Planning
The "Augmented"
LLM
Tools
Memory
Action
Environment

Essential components of AI Agents

🛠️ **Tooling (Chapter II)**
Dynamic access to external resources

💾 **Memory (Chapter III)**
Information persistence & context management

🤝 **Multi-Agent Architectures**
Coordination & collaboration for complex tasks

🎯 **Real-World Applications**
Profitable use cases

📊 **Performance & Observability**
Evaluation, KPIs, error management

Remember These Points

1. Agentivity is a spectrum
Not a binary yes/no—systems have varying degrees of autonomy

2. It's about engineering, not just AI
Success requires robust software architecture and design

3. Hype ≠ Reality
Many "agents" are just workflows; true agents are rare in production

4. The responsibility gap is real
Autonomy raises serious ethical and legal questions

5. Choose frameworks wisely
Trade-offs between power, transparency, and control

Readings for next class

https://arxiv.org/abs/2505.10468



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Full length article

AI Agents vs. Agentic AI: A Conceptual taxonomy, applications and challenges

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https://arxiv.org/abs/2403.15452

What Are Tools Anyway? A Survey from the Language Model Perspective

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Abstract

Language models (LMs) are powerful yet mostly for text generation tasks. Tools have substantially enhanced their performance for tasks that require complex skills. However, many works adopt the term "tool" in different ways, raising the question: *What is a tool anyway?* Subsequently, *where and how do tools help LMs?* In this survey, we provide a unified definition of tools as external programs used by LMs, and perform a systematic review

Final project :

Domain Adaptation of AI Agents

Introduction

AI Agents are now being deployed to automate a growing range of complex tasks — from customer support to document generation.
These agents are built like *puzzles*: each subtask can be solved by a Large Language Model (LLM), but together they form more powerful, autonomous systems.

One of the greatest challenges of *agentic AI* today — as with NLP in general — is **domain adaptation**. Even the largest LLMs perform poorly on very specific or expert-level tasks, and well-formatted domain data is often too scarce for fine-tuning to be practical.
In such cases, **AI agents and innovative agentic architectures** can surpass specialized models — when designed to reason, search, and use external knowledge effectively.

Your mission: **explore how the NLP techniques you know, combined with agent architectures, can solve complex problems that even GPT-5 (or at least the 8B models we'll use) still struggle with.**

The project will be completed in **teams of 2 or 3 students**.
You may choose between two main topics, and then specialize your subject.

Topic 1 : Harnessing Linguistic Ressources with AI Agents,

Annotation projection, Grammatical Rule Mining, Dictionary or Bilingual lexicon generation for low ressource languages

Here is the Lord's Prayer in Old Church Slavonic:

Cyrillic	IPA	Transliteration	Translation
отьчє нашь· иже єси на нєбєсѣхъ: да сѧтитъ сѧ има твоє· да придєтъ цѣсарьствиє твоє· да бѫдєтъ волꙗ твоꙗ ꙗко на нєбєси и на ꙁємли: хлѣбъ нашь насущьнꙑи даждь намъ дьньсь· и отъпоусти намъ длъгы наша ꙗко и мы отъпоущаємъ длъжьникомъ нашимъ· и нє въвєди насъ въ искоушєниє· нъ иꙁбави нꙑ отъ нєприꙗзни:·	otɪtʃe naʃɪ jɪʒe jesi na nebesæxɯ da světitɯ sẽ jɪmẽ tvoje da pridetɯ tsæsaɾɪstvije tvoje da bɔ̃detɯ voʎa tvoja jako na nebesi i na zemʎi. xlæbɯ naʃɪ nasɔ̃ʃtɪnʲjɪ daʒdɪ namɯ dɪnɪsɪ i otɯpusti namɯ dlɯgɨ naʃẽ jako i mɨ otɯpuʃtajemɯ dlɯʒɪnikomɯ naʃɪmɯ. i ne vɯvedi nasɯ vɯ jɪskuʃenije nɯ izbavi nɨ otɯ neprijazni.	*otĭče naší* *Iže jesi na nebesěxŭ.* *Da svętitŭ sę imę tvoje* *da pridetŭ cěsarĭstvije tvoje* *da bǫdetŭ voʎa tvoja* *jako na nebesi i na zeml'i.* *hlěbŭ naší nasǫ̆štĭnyi* *daždĭ namŭ dĭnĭsĭ* *i otŭpusti namŭ dlŭgy našę* *jako i my otŭpuštajemŭ* *dlŭžĭnikomŭ našimŭ* *i ne vŭvedi nasŭ vŭ* *iskušenije* *nŭ izbavi ny otŭ neprijazni.*	Our father Who art in the heavens. May hallowed be thy name may come thy kingdom may become thy will as in heaven, also on Earth. Our supersubstantial bread give us this day and release us of our debts as we also release our debtors, and do not lead us to temptation but free us from the evil one.

Topic 2 : Llama / Qwen 3 passe l'agrégation

Corrigé

Ingenium [ĭngə́nĭŭm] > *engin* [ãžə̨̃]

- Accent : la pénultième [i] étant en hiatus, elle est brève, et l'accent remonte à l'antépénultième [e].
- Les quantités étymologiques peuvent être déduites de la suite de l'évolution vers le français moderne :
  - ainsi, [i] initial est une voyelle brève, puisque la graphie du français moderne montre qu'elle a changé de timbre au moment du bouleversement du système vocalique (à la différence de [i] qui conserve son timbre) ;
  - le déclenchement de la diphtongaison conditionnée indique que la voyelle tonique est un *e* bref [ĕ] devenu [ẹ] (à la différence d'un [ẹ] qui n'aurait pas subi de segmentation).
  - enfin, tout [u] final est bref.

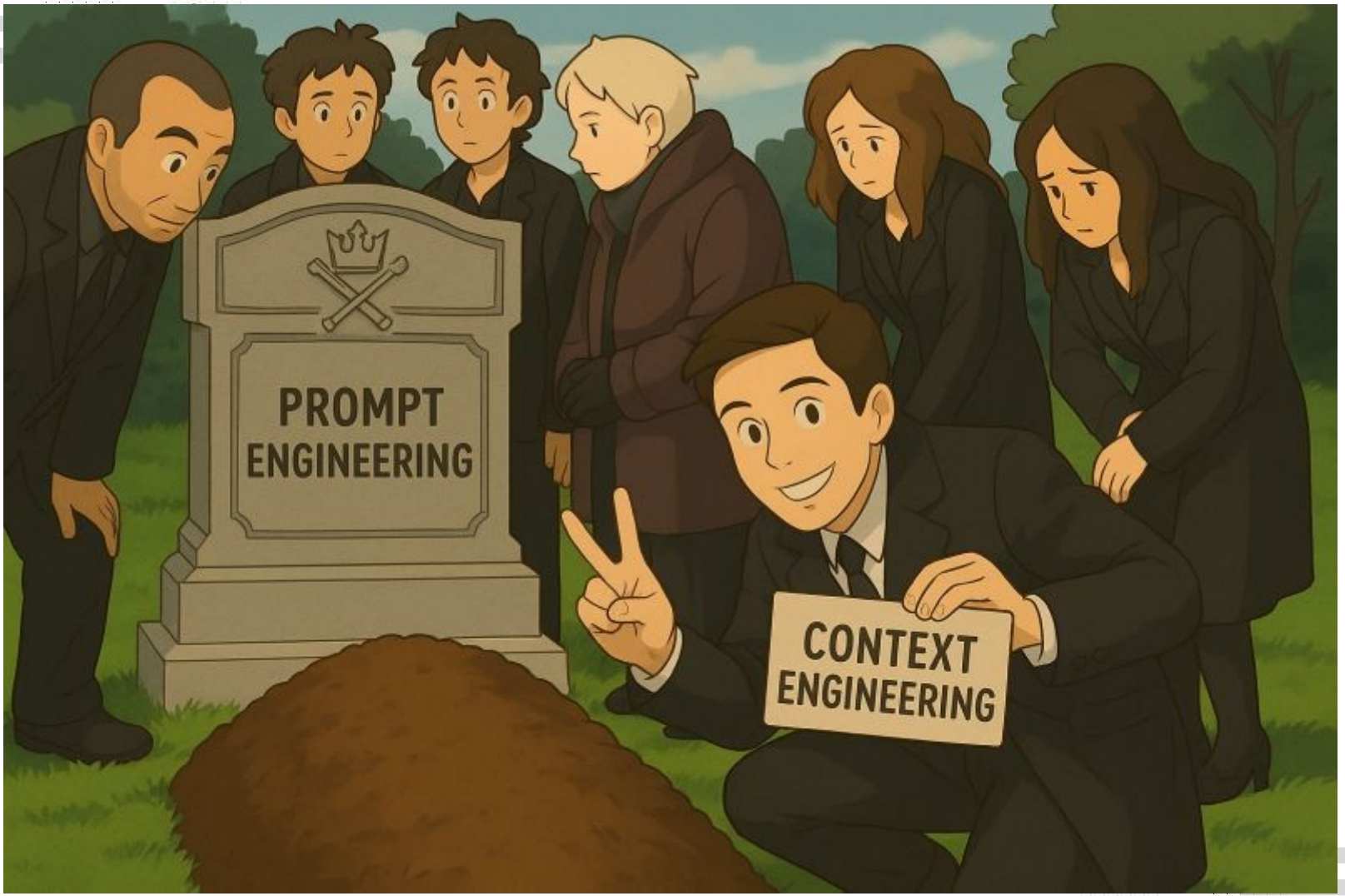
	[ĭngə́nĭŭm]	
I[2] avt	[ĭngə́nyŭ(m)]	– Amuïssement de la consonne finale [m]. – Consonification de la voyelle [ĭ] en hiatus : elle se ferme en semi-consonne [y].
II[e] s.	[ĭngẹ́nyŭ] [ĭngẹ́n̪ŭ]	– Bouleversement vocalique : l'opposition des voyelles selon leur quantité (voyelle longue *vs* voyelle brève) n'est plus pertinente. Elle est remplacée par une opposition d'aperture (voyelle ouverte *vs* voyelle fermée) : [ĕ] > [ẹ]. – Palatalisation de [n] au contact de [y] : recul du point d'articulation (de la zone alvéo-dentale à la zone médio-palatale) et renforcement articulatoire. Ce [n̪] fortement articulé, fait entrave : la voyelle tonique ne peut donc subir de diphtongaison spontanée au III[e] siècle.
III[e] s.	[ẹngẹ́n̪ŭ]	– Suite du bouleversement vocalique : [ĭ] > [e]

	[ẹngẹ́n̪ŭ] [ẹnd̪ẹ́n̪ŭ] [ẹnd̪ž̪ẹ́n̪ŭ]	– Palatalisation de [g] suivi de la voyelle palatale [ẹ] : • en position appuyée, la consonne vélaire avance son point d'articulation jusque dans la zone médio-palatale : [g̪] • puis se dentalise : [d̪] • et s'assibile, c'est-à-dire que se développe lors de la détente, à l'avant de la dentale palatalisée un son chuintant, la constrictive postalvéolaire sonore [d̪ž̪].
IV[2]	[ẹnd̪ž̪ẹ́n̪o] [ẹnd̪ž̪ẹ́ẹn̪o] [ẹnd̪ž̪ẹ́ẹn̪o] [ẹnd̪ž̪ẹ́ẹn̪o]	– Dernière étape du bouleversement vocalique : [ŭ] > [o]. – Début de la diphtongaison conditionnée : [ẹ́] se diphtongue au contact d'une consonne palatalisée subséquente. • au contact de [n̪], la voyelle tonique s'allonge et se segmente : [ẹ́] > [ẹ́ẹ] ; • puis le premier segment se différencie par fermeture de deux degrés successifs, d'abord en [ẹ́], puis en [í].
VII[e] s.	[ẹndžíẹn̪o] [ẹndžíẹn̪o] [ẹndžíẹn̪(o)] [ẹndžíẹi̯n̪] [ẹndžín̪]	– Dépalatalisation de l'affriquée : [d̪ž̪] > [dž]. – Rapprochement des apertures : le second segment de la diphtongue se ferme d'un degré : [íẹ] > [íẹ]. – Chute de la voyelle finale atone. – Désormais implosif (ce qui exige un effort articulatoire supplémentaire), [n̪] développe à l'avant un [i̯] de transition, ce qui facilite la prononciation. Ce [i̯] forme avec la diphtongue une triphtongue par coalescence [íẹi̯]. – Réduction de la triphtongue par coalescence, formée de trois voyelles palatales d'aperture fermée, en deux temps : • assimilation de l'élément médian : [íẹi̯] > [íii̯]; • monophtongaison : [íii̯] > [í]. La forme du texte, *engien*, est une forme dialectale. Selon P. Fouché, elle s'explique par le fait que [n̪], devenu final, « s'est simplement dépalatalisé : il n'y a pas eu formation d'une triphtongue *iey* ni par conséquent de *i* »[2].
XI[e] s.	[ẽndžín̪]	– Nasalisation de la voyelle [ẹ], sous l'influence de la consonne nasale qui suit.
XI[2]-XII	[ẽndžín̪] [ãndžín̪]	– Influence ouvrante de la consonne nasale : en syllabe fermée, la voyelle [ẽ] s'ouvre de deux degrés : [ẽ] > [ẽ] > [ã].
XII[e] s.	[ãndžín]	– Dépalatalisation de [n̪] en position implosive.
XIII[e] s.	[ãnžín]	– Réduction des affriquées, par perte du segment occlusif dental.
XIV[e] s.	[ãnžĩ́n]	– Nasalisation de [i] en syllabe fermée, sous l'influence de la consonne nasale. Cette nasalisation, tardive, a lieu après la date de composition du *Bestiaire*.
XIV[e] s.	[ãnžẽ́n] [ãnžẽ́n]	– À nouveau, influence ouvrante de la consonne nasale : en syllabe fermée, la voyelle *i* s'ouvre d'un degré, puis d'un deuxième : [ĩ́] > [ẽ́] > [ẽ́].
XVI[2]- XVII[1]	[ãžẽ́]	– Dénasalisation partielle : en syllabe fermée, la voyelle garde son timbre nasal et la consonne nasale en position implosive s'amuït.

Expectations:

Code /10
Code an agent for the domain specific task you chose
Gather relevant data
Create tools
Create an architecture adapted to the task
Make your agent accessible to me

Report + article /10 (10 + 6-10 pages)
Create an evaluation datasets
In a report document your experiments and justify your design decisions
Then write a conference like paper containing :
* A SoTA on the task you tackle
* A presentation of the architecture of your agent
* Presentation of its performance and comparison to other solutions


PROMPT
ENGINEERING
CONTEXT
ENGINEERING